Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to CE, EEE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Define the terms Gain, attenuation and decibels. And explain their importance in communications, with examples. [15]

2.a) Draw and describe the various frequency ranges in the electromagnetic spectrum with its applications.
b) Define the sampling theorem. [10+5]

3. Write about the FM Modulation and Demodulation with suitable diagrams and wave forms. [15]

4. Explain about the QPSK modulation and demodulation scheme with suitable diagrams. [15]

5.a) Explain about paging systems with examples.
b) Describe briefly about the Ethernet LANs. [7+8]

6.a) Explain about the Internet Telephony and its applications.
b) Discuss about the Token Ring LAN. [8+7]

7.a) Write in detail about the Global Positioning systems.
b) Explain about the Optical Transmitters and Receivers. [8+7]

8. Explain about the Bluetooth and Zig Bee and write their application in daily life. [15]

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